

3 Tetracycline is a bacteriostatic antibiotic and vancomycin is a bactericidal antibiotic.

(a) Compare and contrast bacteriostatic antibiotics with bactericidal antibiotics.

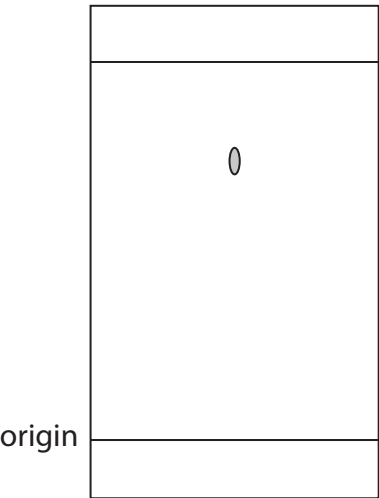
(2)

(b) Antibiotics can be identified using Rf values obtained from chromatography.

The table shows the Rf values for tetracycline and vancomycin, obtained using two different chromatography methods.

Antibiotic	Rf value	
	Method 1	Method 2
Tetracycline	0.51	0.75
Vancomycin	0.09	0.29

The diagram shows a chromatogram for one of these antibiotics using one of the methods.



(i) Describe how this chromatogram could be produced.

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(ii) Which identifies the antibiotic in the diagram and the method used?

(1)

- ☐ **A** tetracycline, using method 1
- ☐ **B** tetracycline, using method 2
- ☐ **C** vancomycin, using method 1
- ☐ **D** vancomycin, using method 2

(iii) Explain why the four values in the table are all different from each other.

(2)

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(Total for Question 3 = 8 marks)